

SHAW CAMPBELL

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Recent Electrical & Computer Engineering Graduate from UCT. Have extensive software development internship experience and machine learning research experience and a good academic track record.

WORK EXPERIENCE



Graduate Software Developer (Internship) – PSG Financial Services

2026 – Present

Building and maintaining production web applications, working across a .NET / Angular stack. Write end-to-end tests using Playwright in C#. Migrated legacy automated tests to Playwright using DeepSeek and OpenCode, requiring a lot of AI engineering.

Skills: C# · .NET · ASP.NET · Angular (Typescript, CSS) · Agile · Web Development · MS SQL Server · OpenCode · Playwright



Software Engineering Vacation Work – Peralex Electronics

2025 · 5 weeks

Designed and built a Python desktop application for visualising and navigating openEMS electromagnetic simulation results across multiple antenna types. Participated in the full Agile workflow – sprint planning, stand-ups, and peer code reviews.

Skills: Python · OOP · Desktop App Development · Agile · Electromagnetics
Reference: Skippy Burger, skippy@peralex.com, [reference letter](#)



Software Development Intern – Electrum Payments

2024 · 4 weeks

Delivered a real feature on a live company codebase, embedded within the engineering team across stand-ups, planning sessions, and code reviews. Received a full-time offer at the conclusion of the internship.

Java · Spring Boot · OOP · REST APIs · Agile
Reference: Johan Du Toit, [Johan Du Toit](#), [reference letter](#)



Electrical Engineering Tutor – Signals and Systems II (EEE3092F)

Jan – Jul 2025 · 4 months

Electrical Engineering Tutor – Signals and Systems I (EEE2047F)

Aug – Nov 2025 · 4 months

Control Systems Laboratory Assistant

2023 & 2024 · 6 weeks per year

Computer Science Tutor – Python (CSC1015F)

2023 · 6 months

Computer Science Tutor – Java (CSC1016S)

2023 · 6 months

SELECTED PROJECTS

Deep Learning Deformable Registration of Ultrasound to MRI for Prostate Imaging

Thesis · 2025

[GitHub](#)

Final-year honours thesis investigating weakly supervised deep learning for deformable registration of transrectal ultrasound (TRUS) and MRI in prostate imaging. Utilised UCT high performance computing cluster, which is run on linux.

Python · PyTorch · VoxelMorph · ANTs · Jupyter Notebooks · Medical Image Analysis · Linux

Micro-Mouse – Sensing & Power Subsystems

Embedded · 2024

[GitHub](#)

Designed and implemented the infrared sensor subsystem and power subsystem for an autonomous maze-solving micro-mouse, interfacing with an existing motherboard and dual-motor drive.

C · STM32L476 · KiCad · Embedded Systems · PCB Design · Electronics

TaskFlow – CPU-Inspired Task Scheduler

C++ · 2025

[GitHub](#)

A back-end C++ task scheduler that adapts classical CPU scheduling algorithms for human productivity planning.

C++ · OOP · Design Patterns · Scheduling Algorithms · CMake

Self Studying Stochastic Calculus

Stochastic Calculus · 2026

Out of love for the rigorous and abstract theory, I have been self studying real analysis and stochastic calculus using textbooks from UCT's masters of financial engineering programme. Progress, handwritten exercise solutions, and a dated study log are available on my [website](#).

EDUCATION



BSc Eng (Hons) – Electrical & Computer Engineering

Honours · 2021 – 2025

AWARDS & ACHIEVEMENTS

- Honours Graduate – GPA = 71%, thesis in medical ML [Honours](#)
- Dean's Merit List 2022 & 2023 [x2](#)
- PSG Full Merit Bursary 2024 & 2025 [Merit](#)
- UCT Plus Gold Award – 60 hours of voluntary community service 2023

EXTRA-CURRICULARS

- Student Residence Mentor**
Jan 2022 – Nov 2023 & Jan 2024 – Nov 2024
Supported first-year students through the academic and social transition into university life across two separate years.
- IT & Maintenance Representative – Leo Marquard Hall House Committee**
Jan 2023 – Nov 2023
Built and maintained a website for the campus residence. Provided IT support across the residence community.

TECHNICAL SKILLS

Python · Java · C++ · C · C# · Linux · .NET · ASP.NET · HTML, JavaScript & CSS · Jupyter Notebooks · OpenCode · MS SQL Server

